Fragile X Syndrome Report

Sample Information

Clinic/Hospital: GENDA S.A.	Referring Doctor: Pilar Igarreta	Referring Department: -
Name: NIRO,MARIA BELEN	Gender: Female	DOB: 1989-09-20
Sample ID: 26B03234389	Sample Type: Blood	Clinic No.: D2667101
Date of Collection: 2026-07-08	Date of Received: 2026-07-13	ID No./Referring No.: -
Clinical Symptoms & Family Medical History:	Family History:- Diagnose:- Symptoms:-	

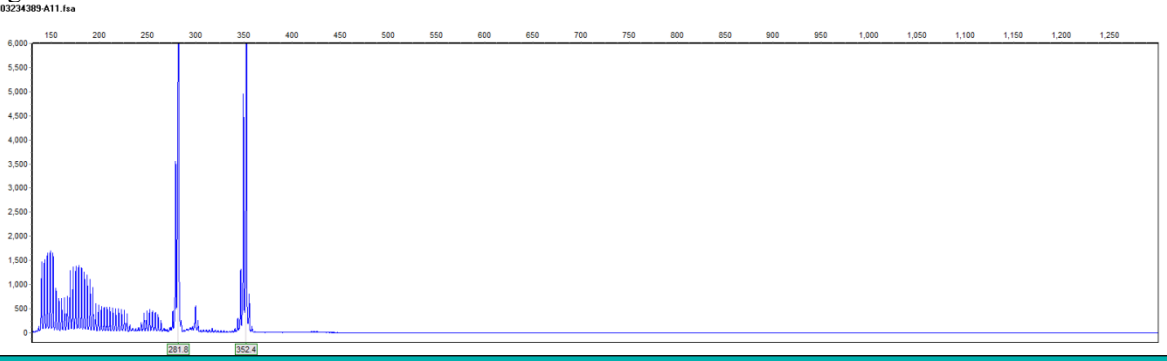
Test Information

Test Item	Fragile X Syndrome Carrier Screening
Test Method	TP PCR and Fluorescent Capillary Electrophoresis

Test Result

CGG Repeats		CGG Repeat Type
Allele1	23	Normal
Allele2	47	Intermediate

Test result figure:



Result Interpretation and Recommendations

The test result shows that the numbers of CGG repeats in the 5' non-coding region of the individual's *FMR1* gene are 23/47, which belongs to Intermediate. Family verification and genetic counseling are recommended.

Note: According to guidelines from the American College of Medical Genetics and Genomics (ACMG) and the "Expert Consensus on Genetic Counseling for Fragile X Syndrome Carrier Screening", the reference ranges for CGG repeats are as follows: normal alleles (CGG repeats < 45), intermediate alleles (CGG repeats: ≥45 to ≤54), premutation alleles (CGG repeats: ≥55 to ≤200) and full-mutation alleles (CGG repeats > 200).

Statement

1. The conclusions above are based solely on laboratory test data, intended for the purpose of mutation detection and provided for clinical reference only. They do not represent the final clinical diagnosis.
2. The reported results apply to the submitted sample only. For any doubts or questions, please contact us within 7 working days upon receipt of the results.

Reviewed by:

Report Date: 2026-07-30

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Disease Overview and Limitations of Testing Methods

Disease Overview

Fragile X syndrome

Fragile X syndrome is a hereditary intellectual disability disorder, encompassing learning difficulties and cognitive impairments. The disease has severe conditions and very poor prognosis. Patients primarily present with varying degrees of intellectual disability, language developmental disorders, autism, distinctive physical features (high forehead, long face, prominent jaw, large ears, loose joints, and enlarged testicles post-puberty), and behavioral abnormalities. The disease has a relatively high incidence rate, affecting 1 in 4,000 males and between 1 in 8,000 to 1 in 5,000 females.

Fragile X syndrome is an X-linked inherited single-gene disorder caused by abnormalities in the *FMR1* gene located on the X chromosome. The gene has a CGG repeat sequence in its 5' Untranslated Region (5' UTR), and there is a CpG island approximately 250bp upstream from this repeat. The instability and expansion of the (CGG)n repeat sequence within the *FMR1* gene, along with abnormal methylation of the CpG island upstream, constitute the molecular basis for this disorder. Approximately 99% of fragile X syndrome cases are caused by CGG dynamic mutations. Based on the range of CGG repeats, the *FMR1* gene can be categorized into normal alleles, intermediate alleles, premutation alleles and full-mutation alleles.

Intermediate alleles in females do not lead to fragile X syndrome, and therefore, prenatal diagnosis is not necessary. However, their offspring may progress to the premutation alleles.

Females with premutation alleles have offspring who are at risk of progressing to the full-mutation alleles, and are therefore required to be subjected to prenatal diagnosis. Furthermore, females with the premutation alleles are at an increased risk of developing fragile X-associated tremor/ataxia syndrome (FXTAS), fragile X-associated primary ovarian insufficiency (FXPOI), and fragile X-associated neuropsychiatric disorders (FXAND). Among them, about 16.5% of females with the premutation alleles will develop FXTAS, and roughly 20% will experience FXPOI. Males with the premutation alleles are at an increased risk of developing FXTAS and FXAND. The risk of FXTAS in males carrying the premutation increases with age, showing a 17% risk of developing FXTAS between the ages of 50 to 59 years, rising to 38% between the ages of 60 to 69 years. escalating to 47% between the ages of 70 to 79 years, and dramatically increasing to 75% over 80 years of age.

Nearly 100% of the males with full-mutation alleles exhibit clinical symptoms, whereas females display variable clinical manifestations depending on the degree of X-chromosome inactivation.

Limitations of Testing Methods

This test employs TP-PCR and fluorescent capillary electrophoresis techniques to analyze the (CGG)n repeats associated with the *FMR1* gene relevant to fragile X syndrome.

This test can rapidly determine the (CGG)n repeats in the *FMR1* gene up to 200 repeats with accuracy, though there may be a discrepancy of ±2 repeats in the result. It is unable to specify the exact (CGG)n repeats in the *FMR1* gene when it exceeds 200 repeats; however, this does not impact the categorization of the allele type.

For the determination of allele types when the CGG repeats are at the threshold values, further assessment in conjunction with clinical context is required.

This test cannot exclude the possibility of sample testing failure due to rare mutations in the primer binding regions.

This test is not capable of detecting methylation abnormalities of the *FMR1* gene, rare point mutations, or other forms of variation.

This test cannot detect AGG insertions within the CGG repeat sequence of the *FMR1* gene.

This test cannot identify samples with full mutation mosaicism below 10% and does not evaluate mosaicism ratios.

Test results may be influenced by the quality of the template DNA, which can be affected by factors such as sample source, collection process, transportation conditions, and sample handling. Additionally, the quality of DNA extraction methods, procedural steps, and limitations of current molecular biology techniques may result in false-positive and false-negative test outcomes.

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Reference

[1] http://omim.org/entry/300624

[2] Hantash FM et al. *FMR1* premutation carrier frequency in patients undergoing routine population-based carrier screening: insights into the prevalence of fragile X syndrome, fragile X-associated tremor/ataxia syndrome, and fragile X-associated primary ovarian insufficiency in the United States. Genet Med. 2011 Jan;13(1):39-45. doi: 10.1097/GIM.0b013e3181fa9fad.
[3] Spector E et al. Laboratory testing for fragile X, 2021 revision: a technical standard of the American College of Medical Genetics and Genomics (ACMG). Genet Med. 2021 May;23(5):799-812. doi:10.1038/s41436-021-01115-y.
[4] Clinical Genetics Group, Chinese Society of Medical Geneticist, Chinese Medical Doctor Association, Clinical Genetics Group, Chinese Society of Medical Genetics, Chinese Medical Association, Group Of Genetic Disease Prevention and Control, Professional Committee of Birth Defects Prevention and Control, Chinese Preventive Medicine Association, et al. Clinical practice guidelines for fragile X syndrome [J]. Chinese Journal of Medical Genetics, 2022, 39 (11):6.
[5] Single Gene Disorder Carrier Screening Consensus Expert Panel, Genetic Counseling Group, Chinese Society of Medical Genetics, Chinese Medical Association, et al. Expert consensus on genetic counseling for fragile X syndrome carrier screening [J] Journal of Reproductive Medicine, 2024, 33 (5): 563-568.

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